From (10)-(12) we obtain

$$\triangle U_4 = U_3 = 0.222\, U_0 \ . \qquad (14)$$

The insulator voltages are unequal. The insulator nearest the line bears the highest voltage and breaks down first. When its RMS voltage satisfies

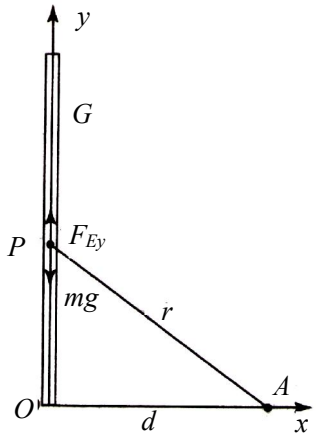
$$\triangle U_1 = U_e \qquad (15)$$

then breakdown damages the string. From (8), (13), (15) the string's maximum withstand voltage is

$$U_{0\max} = \frac{U_e}{0.298} = 369\ \text{kV} \ . \qquad (16)$$

5.

1. As shown, at y, charge P feels repulsive Coulomb force F_E with y component



$$F_{Ey} = k\frac{Qq}{r^2}\sin\theta = k\frac{Qqy}{(d^2+y^2)^{3/2}} \ , \qquad (1)$$

Here r is $P-A$ distance; θ is the $r-x$ axis angle. See F_{Ey} vs y; small y: in (1) numerator y dominates. So F_{Ey} grows with y; large y, the denominator of (1) is dominated by y. Then F_{Ey} decreases with y. F_{Ey} peaks as y increases. To calculate F_{Ey} versus y numerically, set $\tau = y/d$; then

$$F_{Ey} = k\frac{Qq}{d^2}\frac{\tau}{(1+\tau^2)^{3/2}} \ . \qquad (2)$$

Different τ give different $\tau(1+\tau^2)^{-3/2}$; the numerical results are in Table 1.

Table 1

τ	0.100	0.500	0.600	0.650	0.700	0.707	0.710	0.750	0.800
$\tau(1+\tau^2)^{-3/2}$	0.0985	0.356	0.378	0.382	0.385	0.385	0.385	0.384	0.381

The table gives a maximum at $\tau = 0.707$, namely

$$y = y_0 = 0.707d \qquad (3)$$

The maximum of the y component of the Coulomb force is